Neuro-Symbolic AI and Byzantine Law: Rulemapping *Condictiones* of the *Digest*

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Abstract. We investigate the use of neuro-symbolic AI to analyze historical legal interpretations of Justinian's *Digest*. Our case study concerns the Byzantine jurist Stephanus' classification of *condictiones*: personal actions for unjust enrichment. We develop a classifier to investigate how his taxonomy applies to a wider body of *Digest* passages. To this end, we formalize five *condictiones* in propositional logic and operationalize them through Rulemapping, which combines large language model assessments of atomic propositions with logical reasoning. We assemble expert-labelled material to support development and evaluation, providing a benchmark for testing AI's ability to interpret complex legal scenarios. The resulting classifier outperforms a zero-shot baseline on the material used for development, but shows limited generalization when evaluated against classifications drawn from Stephanus' cross-references to other passages. The study illustrates how neuro-symbolic modeling can make historical legal interpretations explicit and testable.

Keywords. Neuro-symbolic AI, Rulemapping, Propositional logic, Benchmark datasets, Computational legal history, Roman law, Byzantine law, *Condictiones*

1. Introduction

Our paper probes AI's legal reasoning capabilities by reference to one of modern law's most challenging sources: the *Digest*, issued by East Roman Emperor Justinian in the sixth century AD. The *Digest* is one of the foundational texts of the Western legal tradition, especially its civil law variant [1]; its way of organizing unjust enrichment, the topic of our study, survives in modern legal orders to the present day, most notably § 812 of the German Civil Code [2]. But the *Digest* holds promise for AI and law research as well. Most importantly, it exhibits the same structural features as precedents, which large language models (LLMs) find difficult to parse [3]. Not only does the *Digest* combine restatements and applications, facts and hypotheticals; it also makes a point of enshrining differences in opinion. As such, it constitutes a probative corpus on which to test AI's abilities to find, or make, legal meaning in the face of multivalence and contradiction.

Meanwhile, the *Digest* continues to be of interest to legal historians. But the vastness of the *Digest*'s scale, the heterogeneity of its sources, and the multiplicity of its authors pose challenges for conventional approaches to ancient legal texts. As a consequence,

scholars of the *Digest* have long been interested in computational research technologies. Early projects such as the ROMTEXT database transformed the text of the *Digest* into a machine-readable corpus and enabled new forms of computerized search and analysis [4]. Yet the possibilities of computational research extend beyond textual retrieval; digital tools can offer legal historians "new ways of seeing data, patterns, and relations" [5]. Building on a relational *Digest* database developed to support structured queries [6], Ribary and McGillivray used hierarchical clustering to investigate the text's conceptual structure and word embeddings to explore differences between general and legal Latin usage [7].

In sum, we see that the *Digest* is of interest to researchers of both AI and of legal history: on the one hand, as a domain for benchmarking AI performance on complex legal reasoning tasks; and on the other, as a corpus for computational legal-historical analysis. These interests motivate the present work, in which we develop and evaluate a neuro-symbolic system that classifies passages from the *Digest* according to the taxonomy of *condictiones* developed by the sixth-century Byzantine jurist Stephanus. *Condictiones* are personal actions that could be brought in cases of unjust enrichment, and identifying which of them applies in a situation requires a careful analysis to select the right interpretation from multiple possible readings of at times complex scenarios. From a legal-historical perspective, such a classifier could offer a means of extrapolating Stephanus' reasoning beyond those *Digest* fragments for which his commentary survives. This, in turn, could help scholars assess the scope and limits of his taxonomy.

We build this classifier with the neuro-symbolic Rulemapping method [8,9], which operationalizes a propositional logical specification through the use of LLMs. Logical modelling has a long history in AI and law because legal text often follows the kind of structure captured by formal logical connectives [10,11]. Just as often, however, legal language involves vague and context-dependent concepts that resist a simple logical description. LLMs' are promising tools for handling such concepts, but their use comes at the cost of determinacy and explainability [12]. Rulemapping aims to combine the strengths of these approaches: LLMs assign truth values to atomic propositions, while a logical reasoning component determines the consequences of those assignments.

This paper makes three contributions. First, we formalize in propositional logic the five types of *condictiones* distinguished by Stephanus in the first title of the twelfth book of the *Digest* (*D*. 12.1); cf. Table 1. Second, we assemble expert-labelled material comprising 25 fragments from *D*. 12.1 and ten distinct fragments elsewhere in the *Digest* identified through Stephanus' cross-references. This material provides a benchmark for testing AI's ability to interpret legal scenarios with multiple, potentially conflicting readings. Third, we operationalize the formalization through Rulemapping, using iteratively refined prompts for `GPT-5.5` [13] to assess whether the constituent propositions hold for the situations described in the fragments. We use the *D*. 12.1 material to develop the classifier and the cross-reference material to evaluate its transfer beyond that title. We also apply the classifier more widely across the *Digest*, exploring how Stephanus' taxonomy might extend to fragments for which no commentary by him survives.

The paper is structured as follows. Section 2 introduces the historical background, develops a logical formalization of Stephanus' taxonomy, and explains how it is operationalized through Rulemapping. Section 3 describes dataset construction and the experimental setup, and presents quantitative and qualitative evaluations of the classifier, both on *D*. 12.1 and beyond. Finally, Section 4 discusses the implications of our findings for computational legal history, and identifies directions for further refinement and research.

2. Methodology

In this section we describe the method by which we analyzed Stephanus' classification of the *condictiones* in the *Digest*. We start in Section 2.1 with a primer on related historical background to explain what the *condictiones* are and where they came from. In Section 2.2 we give a propositional-logical formalization of the conditions under which the *condictiones* of *D*. 12.1 apply. Lastly, we explain in Section 2.3 how this formalization can be operationalized by the neuro-symbolic version of the Rulemapping method.

2.1. Background: The Digest *and its* Condictiones

In the divided Western and Eastern Roman Empire, Emperor Justinian saw legal fragmentation as a cause of political disunity and tasked a commission with creating a code based on classical Roman law. The commission condensed and clarified valid laws into the *Digest*, a fifty-book Latin compilation covering civil matters such as contracts, property, marriage, and wills. Title *D.* 12.1, the name of which translates to "Things credited giving rise to fixed claims and the *condictio*," lies at the heart of our study. A *condictio* is a personal action in which the plaintiff demands the return of something. In the *Digest*, *condictiones* are referred to only in general terms—typically as forms of the remedy *condictio* or as derivatives of the verb *condico*—without further specification. Substantively, it is difficult to identify a single unifying basis for the various applications of the *condictio*; unjust enrichment seems to be the only common denominator underlying its use.

Stephanus taught Justinian's codification by translating and explaining the Latin text of the *Digest* in Greek. Fragments of these teachings are handed down in the old scholia to the *Basilica* (893) [14]. Stephanus sought to make the principles guiding the application of the *condictiones* explicit in his teaching, by distinguishing and naming specific types of *condictiones* within *D*. 12.1 [15]. His systematic approach served a clear pedagogical purpose. He identified the following categories (rendered in Latin): *condictio certi generalis* [16], *condictio certi*, *condictio ex mutuo*, *condictio de bene depensis* [17], and *condictio ex bono et aequo* [18]. Aside from *D*. 12.1 the *Digest* contains seven other titles concerning *condictiones*, but only a limited number of fragments have survived in which Stephanus refers to those other titles.

2.2. A Logical Formalization of the Condictiones

In this section we describe Stephanus' *condictio* taxonomy and formally define it using standard propositional logic; see Table 1. We note from the onset that this formalization was made with the pragmatic purpose of classifying *condictiones* in mind. For this reason, we make some formal distinctions between *condictiones* which were not made by classical jurists. We will point out these differences below, where applicable.

Condictio ex Mutuo. The *condictio ex mutuo* is a prototypical *condictio* because it represents the original use of the *condictio* in *D*. 12.1 [15], and as such clearly showcases the role of ownership, which we denote by *O*. *Condictio ex mutuo* roughly means "*condictio* based on a *mutuum*," where a *mutuum* denotes a loan of fungible goods. The typical example of such a good is money, and for the sake of simplicity we will restrict our discussion in this section to situations involving money. There are two requirements for a *mutuum* to come about: there needs to be consent between the parties involved (*consensus*)

Table 1. Our propositional logical formalization of Stephanus' taxonomy of the *condictiones* in *D*. 12.1.

Type	Symbol	Meaning	Definition
Atomic proposition	O	Plaintiff proceeds as owner	—
	D	*Datio*: ownership transferred	—
	S	Consensus on the loan	—
	U	*Consumpta*: consumed in good faith	—
	K	*Certum* claimed	—
	L	Loan claim at issue	—
	T	Enrichment through third party	—
	PS	*Pecunia soluta*	—
	IC	*Ex iniusta causa*	—
	LC	*Post lustrum completum*	—
	VF	*Vi fluminum importata*	—
	CF	*Ex causa furtiva*	—
Formula	M	*Mutuum*: loan for consumption	$D \wedge S$
	G	Possession without valid cause	$PS \vee IC \vee LC \vee VF \vee CF$
	CM	*Condictio ex mutuo*	$\neg O \wedge M \wedge L$
	CBD	*Condictio de bene depensis*	$\neg O \wedge \neg M \wedge U$
	CBA	*Condictio ex bono et aequo*	$\neg O \wedge \neg M \wedge T$
	CC	*Condictio certi*	$\neg O \wedge K$
	CCG	*Condictio certi generalis*	$O \wedge G$

and there needs to be a transfer of ownership (a *datio*) [15]. Respectively denoting these propositions by the letters S and D, we may thus define a *mutuum* M as the conjunction $M = D \wedge S$. The idea behind a *mutuum* is that the money is lent in order to be spent. As such, the lender does not expect to receive the exact coins that he lent to be returned; the lendee now owns the money, and must eventually repay the debt. Failing to do so, the lender may bring the *condictio ex mutuo* to demand repayment. Letting L denote such a claim, we formalize this *condictio* by the formula: $CM = \neg O \wedge M \wedge L$.

Condictio de Bene Depensis. The next *condictio* we consider is called the *condictio de bene depensis*: "the *condictio* concerning things spent in good faith." It was intended to govern scenarios in which fungible goods were transferred while the conditions for a *mutuum* failed: either there was no consensus ($\neg S$), or there was no transfer of ownership ($\neg D$). An example is given in *D*. 12.1.13pr. by Ulpian: a thief who pays coins with the intention of making a loan does not transfer ownership ($\neg D$), because he has no authority to do so. However, unaware of the failed *mutuum*, the receiver may then proceed to spend the money in good faith, but because there was no *mutuum* the *condictio ex mutuo* does not apply. On the other hand, ownership had now passed ($\neg O$) because the money was spent by the receiver. It is important here that the money was spent in good faith, which we denote as proposition U. We thus define this *condictio* by: $CBD = \neg O \wedge \neg M \wedge U$.

Condictio ex Bono et Aequo. Next is the *condictio ex bono et aequo*: "the *condictio* based on what is good and equitable." It applies to situations in which the defendant obtained the money from the plaintiff in good faith through another party to whom the plaintiff was not contractually bound [15]. We denote this status of the defendant as a third party by the proposition T. The example of such a situation given by Celsus in *D*. 12.1.32 is that the debtor of the plaintiff lends money to the defendant, while the defendant mistakenly believes that the money is lent to him by another person's debtor. As such there is no consensus on the loan ($\neg S$), and therefore there is no *mutuum* ($\neg M$). The formula for this *condictio* is with the involvement of a third party: $CBA = \neg O \wedge \neg M \wedge T$.

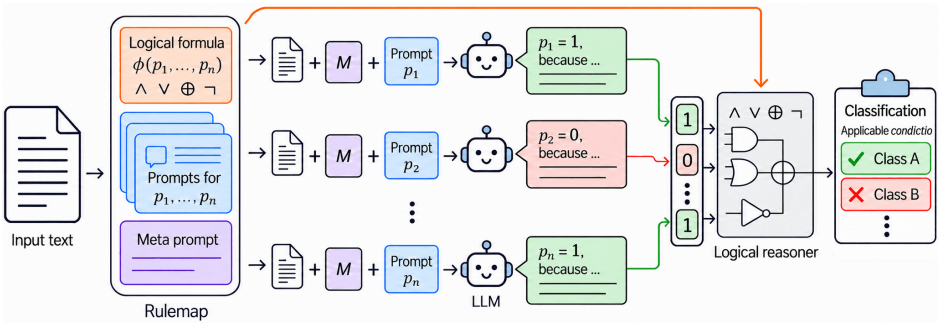


Figure 1. Visualization of the Rulemapping classifier. The Rulemap consists of the formula that defines the logic of the classification task, together with the proposition prompts and the meta prompt which the LLM uses to process the input text to determine the truth values of the propositions $p_1,\ldots,p_n$. Its outputs are substituted in the logical formula and handed to the logical reasoner which subsequently computes the output.

Condictio Certi (Generalis). Lastly we consider the *condictio certi* and the *condictio certi generalis*. A *certum* means "something definite," so *condictio certi (generalis)* roughly means "the (general) *condictio* of something definite." These *condictiones* stand somewhat in contrast with the first three we discussed. Firstly, the *condictio certi* is a subcategorization of the *condictiones* which was already present in the *Digest* while the *condictio certi generalis* was part of Stephanus' classification and did not appear under that name in the *Digest*. Secondly, they are defined based on the nature of the thing that is to be returned—a *certum*—rather than in terms of the circumstances in which they apply. We denote the *certum* claim by the proposition K. We further specify the cause of this situation, denoted by G, as a disjunction of the various circumstances under which O might apply: money paid in connection with a pledge (PS), fruit received without a legal basis (IC), the completion of a five-year period (LC), land accreted by the river (VF), or a thing obtained from a theft-related cause (CF—as in the example above). We distinguish these *condictiones* on the ownership question: $CCG = O \wedge G$ and $CC = \neg O \wedge K$.

As such, these *condictiones* conceptually overlap to a large degree with the three previously discussed. In his writing, Stephanus preferred to apply the more specific *condictio* to a scenario: if both a *condictio certi* and a *condictio ex mutuo* apply, then he would discuss the latter. We follow this approach in our formalization, and define the *condictio certi (generalis)* in such a way that the more specific *condictio* is applied first.

2.3. The Rulemapping Method

We use the neuro-symbolic variant of Rulemapping [8,9] to create a system that is capable of classifying which *condictio* applies to a situation described in a piece of input text. Rulemapping is a general method for creating classifiers that combine a propositional logical reasoning component with an LLM. Such a classifier consists of (1) a propositional logical formula $\phi(p_1,\ldots,p_n)$ built from n atomic propositions $p_1,\ldots,p_n$, (2) an LLM prompt Prompt(p_i) for each proposition p_i, (3) an LLM meta prompt M, (4) a choice of LLM, and (5) a logical reasoning component. We refer to components (1)–(3) as the *Rulemap*, and to components (1)–(5) simply as the *system* or the *classifier*; see Figure 1.

The formula $\phi(p_1,\ldots,p_n)$ of the Rulemap breaks a rule such as a legal norm into its constituent propositions, using the connectives of propositional logic. The root of the

formula poses the overarching legal question, and the propositions $p_1, \ldots, p_n$ correspond to the legally relevant questions pertaining to that rule. We refer to this formula ϕ as the *external* logical structure of the Rulemap. For our use case, we use the disjunction $\phi(O,D,S,...,CF) = CM \vee CBD \vee CBA \vee CC \vee CCG$ from Table 1. A Rulemap also comes with prompts of two types: a meta prompt, and for each proposition p_i a prompt Prompt(p_i) that defines it: the question it poses, its *ratio legis*, and the criteria that must be met in order for the proposition to be satisfied (e.g., routes to satisfaction, defeaters, and subcriteria). When these criteria are of a logical nature then we refer to them as the *internal* logical structure of the Rulemap, because it is handled by the LLM and is not part of ϕ.

Given a Rulemap, classification of a case-description input proceeds in two steps. First, for each proposition p_i the LLM is given the input text together with the meta prompt M and the proposition's prompt Prompt(p_i). It then outputs a truth value $b_i \in \{0,1\}$ for p_i together with a short text justifying this choice (notwithstanding the possibility that this justification may not be entirely faithful [19]). The b_i truth values are then substituted in ϕ to yield $\phi(b_1, \ldots, b_n)$, which is then evaluated and used to produce the system's output classification by the logical reasoning component.

We note that this presentation of Rulemapping was simplified for the sake of clarity. The actual software is more sophisticated: it uses lazy evaluation for propositions, adds data pre- and post-processing steps between the engine and the LLM, is capable of using fuzzy logic instead of propositional logic, and provides a visual interface for development, testing, and execution of Rulemaps.[1] We refer the reader to [9] for further details.

3. Classifying *Condictiones* in the *Digest* with Rulemapping

In this section we use the formalization of Table 1 to create a Rulemapping classifier and run this on the *Digest*. In Section 3.1 we describe technical details of the experiment. We then evaluate the performance of the Rulemapping classifier in terms of quantitative measures in Section 3.2. Then, we evaluate it qualitatively in Section 3.3 by means of a more in-depth comparison between some of the classifications by the system, the rationale it provided for those classifications, and the assessment provided by Stephanus.[2]

3.1. Dataset Construction and Experimental Setup

We obtained the *Digest* from the Grenoble *Corpus Iuris Civilis*,[3] split the books into fragments at the subsection markers, and filtered by keyword search for every fragment matching the stem of *condicere*. This yielded 328 fragments across 37 of the 50 books. Within such a fragment, the model must identify the decisive case among potentially competing scenarios and interpret Latin against English criteria and German engine tags. A domain expert, Hylkje de Jong, prepared two gold standards from Stephanus' classifications. The first comprises 25 labelled fragments from *D.* 12.1, which we split randomly into a training set of twelve and a held-out set of thirteen. The second consists of Stephanus' cross-references from the *Basilica* scholia to *condictio* cases outside *D.* 12.1. An overview of these references is provided in Table 2.

[1] Rulemaps can be built and reviewed at `https://builder.rulemapping.org/`.

[2] All code and data related to the experiment are available at `https://github.com/ovcsl/byz_con`.

[3] We used `https://droitromain.univ-grenoble-alpes.fr/`, though this resource is no longer publicly available at the time of writing. There are alternatives such as: `https://www.iurisprudentia.de`.

Table 2. A comparison of ten cross-references by Stephanus to *Digest* books outside *D.* 12.1. Each row lists the origin of the reference in the *Basilica* scholia (*BS*), the *Digest* fragment it points to, the *condictio* assigned by Stephanus, the *condictio* assigned by the system, and the propositional letters on which the two accounts differ (if any). Multiple references to the same fragment are listed only once, resulting in 6 omissions.

Scholion	Fragment	Stephanus	System	Proposition
BS 364/31–365/21	*D.* 2.15.3.2	*CBA*	*CC*	*T*
BS 1495/11–1496/25	*D.* 13.3.1pr.	*triticaria*	*CC*	—
BS 1503/1505/5	*D.* 16.3.13.1	*CCG*	*CC*	*O*
BS 1503/1505/5	*D.* 25.2.25	*CCG*	*CCG*	—
BS 1527/25–1528/3	*D.* 44.7.24pr.	*CBD*	*CBD*	—
BS 1534/21–1535/8	*D.* 14.6.9.1	*CCG*	*CBA*	*O*, *T*
BS 1543/6–18	*D.* 12.4.12	*CCG*	*CC*	*O*
BS 1555/18–27	*D.* 19.1.24.1	*CBA*	*CCG*	*O*, *T*
BS 3302/8–3303/11	*D.* 13.1.4	*CBA*	*CBA*	—
BS 3302/8–3303/11	*D.* 15.1.3.12	*CBA*	*CBA*	—

We developed the Rulemap with `GPT-5.5` until all twelve training fragments were classified correctly. This version agreed with Stephanus on eleven of the thirteen held-out fragments. With the domain expert, we then revised the prompts to address the remaining errors, reaching agreement on all 25. This final configuration was frozen and run once over the full corpus, including the cross-references.

We ran all experiments on `GPT-5.5` (unless stated otherwise) at temperature 0.1, top-p 10^{-5}, a fixed seed and extended reasoning turned off, so decoding is near-deterministic. On the labelled set of *D.* 12.1 the configurations of `GPT-5.5` and `GPT-4o` [20] were each run five times, giving 125 classifications per configuration; the runs on the cross-references and on the full corpus were only done once. Each classification is archived together with the verdict and justification of every leaf that contributed to it in the GitHub repository.

3.2. Quantitative Evaluation

We now discuss the outcomes of the experiments in quantitative terms: we compare the Rulemapping method against a zero-shot baseline, and discuss some of the statistics of the comparison of the classifier against the gold standard set by Stephanus on the *Digest*.

Zero-Shot Baseline Comparison. To evaluate the influence of the logical scaffolding we ran the 25 fragments of *D.* 12.1 in a zero-shot baseline condition, in which a standalone LLM receives the fragment and the five *condictiones* with Stephanus' definitions but without the Rulemap, i.e. without its formula, prompts, or meta-prompt. Every configuration was repeated five times; Table 3 reports the majority vote over those repetitions, balanced accuracy with a bootstrap interval, and Krippendorff's α values.

Without the Rulemap, `GPT-5.5` comes to the right conclusion in 5 out of 25 fragments; with it, all 25 fragments are classified correctly. Unaided, neither `GPT-5.5` nor `GPT-4o` outperform the baseline of guessing the largest class (36%); the gap between them appears only once the Rulemap scaffold is in place. Additionally, the Rulemap raises accuracy as well as self-consistency, from $\alpha = 0.71$ to $\alpha = 0.98$, meaning that repeated runs of the same configuration agree almost always. However, we should qualify this comparison by noting that the proposition prompts of the Rulemap were tuned on the

Table 3. Classification statistics of the Rulemapping classifier and a zero-shot baseline on the 25 labelled fragments of *D*. 12.1, five repetitions each. Balanced accuracy is reported due to class imbalance. The Krippendorff's α coefficients measure agreement between repetitions of the same configuration.

Model	**Mode**	**Majority**	**Bal. acc.** [95% CI]	α
`GPT-5.5`	Rulemap	25/25 (100%)	98.9 [97.5–100.0]	0.98
`GPT-5.5`	Zero-shot	5/25 (20%)	61.4 [54.0–67.5]	0.71
`GPT-4o`	Rulemap	16/25 (64%)	69.5 [62.5–80.8]	0.72
`GPT-4o`	Zero-shot	8/25 (32%)	60.9 [48.3–72.6]	0.70

training set with the scaffolding in place, while we did not attempt to further finetune the zero-shot prompts.

Transfer to Stephanus' Cross-references. In his writing on the *condictiones* Stephanus refers a number of times to *Digest* fragments outside *D*. 12.1 in relation to this classification. By means of testing the performance of our classifier on the *Digest* books outside *D*. 12.1, we collected all such references and compared them to the classifications made by our system. In total, there are ten (non-duplicate) such references; see Table 2.

Stephanus references one fragment as relating to the *condictio triticaria*—the counterpart of the *condictio certi* for determinate things other than money—which did not occur in *D*. 12.1. Because of this, the *condictio triticaria* is not part of the system's Rulemap, and therefore it is unable to output this *condictio* as its classification, regardless of the case description. Instead, the system returned the *condictio certi*, which is identical to the *condictio triticaria* modulo the type of good of which ownership is claimed. As such this classification is essentially correct. Modulo the *triticaria* mismatch the system thus agrees with Stephanus on 5 out of 10 of the cross-references.

All disagreements are based on two letters: O, which fails in 4/5 misclassifications; and T, which fails in 3/5 misclassifications. As such, we can conclude that in particular the ownership proposition, which is of central importance to *condictiones* in general, did not generalize well enough to books outside of *D*. 12.1. Because the ownership is taken as an atomic proposition in the Rulemap, the logical reasoning related to this proposition is left entirely to the LLM (i.e., is part of the internal logic). Additionally, it is the proposition which required the most finetuning during the training stage. Its performance on *D*. 12.1 became adequate only after the addition of a few principled case instances to the proposition's prompt. We suspect that the LLM became overly reliant on these instances for its classifications, and as such failed to generalize to the other books.

Evaluating the Full Corpus. We ran the final version of the classifier once over the extracted corpus: 328 fragments containing an explicit *condictio*-form across 37 books of the *Digest*. Beyond the references of Table 2, no gold standard covers this material, so these results describe the system's assignments rather than their accuracy.

Most fragments (55%) were classified as the *condictio certi* by the system. This is to be expected because this *condictio* acts as a default category when no more specific *condictio* applies: $CC = \neg O \wedge K$. A further 23% remained "unclassified," because none of the five *condictiones* applied under the Rulemap. This may be due to misclassification or to fragments describing types of *condictiones* that are not amongst the five discussed in *D*. 12.1. For example, of the 70 *condictio*-fragments in *D*. 12.4 to 12.7, 41 receive *condictio certi* and 26 remain unclassified. These titles concern *causa data causa non secuta*, *ob turpem causam*, *indebiti* and *sine causa*, none of which is part of the Rulemap.

3.3. Qualitative Evaluation

A feature due to the use of LLMs in the Rulemapping classifier is that every *condictio* classification is accompanied by generated justifications for why the relevant propositions were deemed to apply or not, which provides some insight into how the system reasons. On the other hand, previous research has indicated that such justifications may not accurately reflect the internal workings of the LLM [19]. Practically, however, they are quite useful: Rulemaps are constructed by means of a back-and-forth between running the classifier, inspecting the LLM's justification, and tuning the Rulemap accordingly.

In this section, to get a better understanding of the performance of our classifier, we use the models' justifications to qualitatively evaluate a particular discrepancy between Stephanus' classification and that of the system: conflating the *condictio certi generalis* and the *condictio ex bono et aequo* in complex cases. For *D.* 13.1.4, Stephanus and the system agree on the *condictio ex bono et aequo*. For *D.* 19.1.24.1, however, Stephanus uses the *condictio ex bono et aequo*, while the system uses the *condictio certi generalis*.

First Comparison: D. *13.1.4.* In relatively straightforward cases, our system accurately replicated Stephanus's classification; *D.* 13.1.4. exemplifies this. At the hands of Justinian's expert commission, third-century jurist Ulpian's assertion that "[i]f a slave or a son under paternal control commits theft, a *condictio* will go against the master for whatever has come to him…" entered the *Digest* as *D.* 13.1.4. As noted above, the *Digest* did not specify which kind of *condictio* applied to the case at hand, but Stephanus offered clarification: "The owner is liable… with the *condictio ex bono et aequo* in case of theft. Because it is not just that the owner of the slave would gain a profit to the detriment of another…" Due to the fact that Ulpian's fragment easily satisfies the antecedent of $\neg O \wedge \neg M \wedge T \rightarrow CBA$, our system correctly classified *D.* 13.1.4 as an instance of the *condictio ex bono et aequo*. It generated the following justification for T:

> … The claim is limited to what has come to the master, showing identifiable enrichment in his assets from property taken from the claimant, with no contractual relation concerning that thing. This is a third-party enrichment constellation; the criterion is satisfied.

Second Comparison: D. *19.1.24.1.* In a more complicated case, however, our system failed to accurately replicate Stephanus's classification. The case in question concerned a situation in which a third party, acting in good faith, had purchased a slave stolen from his master by a thief. The slave in question then used his master's *peculium*, assets given to him to administer, to buy a second slave for his new master. At issue was whether the original master of the first slave had a *condictio* against the good faith purchaser in relation to the second slave bought with the *peculium*. The *Digest* treated this case as follows: "Sabinus said that you can bring a *condictio* against me for this slave, but that if I lost anything in the transaction which he arranged, I will have a counteraction on the *peculium*." (*D.* 19.1.24.1). In a remark on *D.* 12.1.31.1, which concerned a case similar to that of *D.* 19.1.24.1, Stephanus posited that the relevant *condictio* in this case was the *condictio ex bono et aequo*. After all, a third party (purchaser) had been enriched at the expense of the original owner (owner of the first slave), without any contractual or quasi-contractual relationship having come into existence between them.

In contrast, our system reasoned that the first slave's original owner had automatically become the owner of the second slave as well—which is to say, that O applies in this case:

> ... The claim is tied to the fact that the acquisition was made from the master's *peculium*, so the purchased slave is treated as belonging to him by operation of law; the *condictio* functions to recover that specific slave. This is not merely a personal debt or delictual *condictio*. Conclusion: the criterion is satisfied.

Additionally, it correctly stated that *G* is justified here as *ex causa furtiva*, and so it judged that the *condictio certi generalis* can be brought by the plaintiff. However, *O* should not apply to this case: while it is true that the plaintiff remained owner of the first slave because it had been stolen, the *condictio* is brought here with respect to the second slave, of whom, according to Stephanus, the plaintiff apparently had never been the owner.

4. Discussion and Conclusion

In this paper, we formalized Stephanus's classification of the five *condictiones* discussed in *D*. 12.1 in propositional logic and operationalized it through neuro-symbolic Rulemapping. We assembled expert-labelled material based on Stephanus's commentary to develop and evaluate the resulting classifier. This material also provides a benchmark for testing AI's ability to interpret complex legal scenarios. After prompt refinement, the final `GPT-5.5` configuration correctly classified all 25 labelled fragments of *D*. 12.1, compared with five in the zero-shot configuration. However, agreement with Stephanus's cross-references beyond *D*. 12.1 was lower, indicating limits of the classifier's ability to generalize. The evaluations showed that these difficulties primarily relate to the ownership proposition *O*. Decomposing *O* into more specific propositions connected by the Rulemap's external logic would transfer part of the ownership reasoning from the LLM to the symbolic component. Rulemapping's advantage over zero-shot classification in our baseline comparison (Section 3.2) suggests that this transfer could improve performance in future work.

To fully realize the potential of this approach, however, the Roman-law rules underlying the method still require further refinement. For example, in cases involving theft (such as *D*. 19.1.24.1), the rules specifying third-party involvement need to be refined, in order to clearly distinguish between the *condictio ex bono et aequo* (Stephanus) and the *condictio certi generalis* (LLM). The same goes for the interpretation of the relationship between an owner and his slave and between a father and his son, particularly the absence of a general concept of representation in Roman law and the mechanisms that created additional liability for contracts entered into by a slave or son under paternal control on behalf of the owner or father. Through continued collaboration with domain experts these dependencies could be made explicit and incorporated into the Rulemap.

More generally, our findings also suggest the expansion of AI modeling to other Roman and Byzantine legal actors and sources. Structure, organization, and impact of the *Digest* are famously opaque and, in spite of ongoing attention from the literature, many questions remain. Using computational methods to juxtapose how earlier and later legal scholars dealt with the same *Digest* passages could provide insight into how the same provision came to be understood differently across time and place. At the same time, applying these methods opens up new avenues of inquiry by making it possible to pose questions that were previously difficult, if not impossible, to formulate. In particular, computational approaches could enable scholars to investigate semantic questions across large bodies of data, identifying patterns and shifts in meaning on a scale previously impractical or attainable only through years of manual research.

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